Aditya Kumar Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.05
Delhi Public School, Ranchi	Class XII (CBSE)	2024	94.80%
Bridgeford School, Ranchi	Class X (CBSE)	2022	98.20%

SCHOLASTIC ACHIEVEMENTS

- Awarded the '**Merit Certificate**' for scoring **100% in Maths**, **school topper** in Class X CBSE.
- Achieved a **maximum rating of 1535 (Specialist)** in Codeforces (*Codeforces Profile*).
- Achieved a **maximum rating of 1736 (3 star)** in Codechef (*Codechef Profile*).
- Solved **more than 900 problems** across Codeforces and Leetcode.

SKILLS

- **Languages** -- C, C++, Python, Dart, JavaScript, SQL, HTML/CSS
- **Frameworks/Libraries** -- Flutter, Node.js, Express.js, React, Socket.io, fl_chart, Monaco Editor, Yjs, BullMQ
- **Technologies** -- PostgreSQL, Redis, Docker, Git/GitHub, GitHub Actions (CI/CD), Railway
- **AI-Assisted Development** -- Using LLM coding assistants (Claude Code, Copilot) for debugging, code review, and rapid prototyping
- **CS Fundamentals** -- Data Structures & Algorithms, OOP, Operating Systems

RELEVANT COURSES

- **Mathematics**: Introduction to Probability, Transform Techniques, Elementary Linear Algebra, Differential Equations, Complex Variables, Introduction to Statistics, Calculus - I & II.
- **Computer Science**: Introduction to Programming, Data Structures & Applications

PROJECTS

Collaborer - Real-Time Collaborative Code Editor | *Personal Project* | [*GitHub Link*] | [*Live Link*]

- Built a real-time collaborative code editor using Yjs CRDTs to **synchronize concurrent edits across distributed clients** -- automatic, conflict-free merging with no central locking.
- Integrated **Monaco Editor** with multi-language syntax highlighting, using WebSocket-based sync to maintain consistent state across simultaneous multi-user sessions.
- Enabled in-browser **code execution** across multiple languages via the **Glot.io API**, routing submissions to the appropriate language runtime and streaming back stdout/stderr without requiring any local environment setup.
- Deployed on **Render**, managing persistent WebSocket connections and **concurrent session state** across multi-user editing rooms.

JudgeSync - Collaborative Competitive Programming Platform | *Personal Project (In Progress)* | [*GitHub Link*]

- Built a sandboxed **multi-language execution engine** (C++, Python, Java, JS) with process isolation, timeout enforcement, and verdict classification for safe concurrent execution of untrusted code.
- Architected a distributed job pipeline using **BullMQ/Redis** to decouple API handling from execution, enabling **concurrent worker execution** and **isolating request latency** from compute-heavy runs.
- Developing a **React frontend** delivering real-time verdicts **via Redis pub/sub over Socket.io**, coordinating state across concurrent client connections without polling.

NexRep - Flutter Fitness Tracking App | *Personal Project* | [*GitHub Link*]

- Architected a **custom singleton-based state system** using **ChangeNotifier**, avoiding heavier state management overhead for persistent timer sessions across navigation.
- Built a **multi-chart analytics engine** computing 1RM (Epley formula), volume trends, and muscle-group distribution in real time.
- Designed an **offline-first persistence layer** with on-device aggregation logic using **shared-preferences** with zero network calls.

POSITION OF RESPONSIBILITY

- **Head Internship Coordinator of MAE, OCS** - Office of Career Services
- **Core Member, Infero** - Competitve Programming and Logical Reasoning Club
- **Events Head** - Mechanical Department Day, 2026 -- led cross-functional coordination across logistics, sponsorships, and student teams.
- **Teaching Assistant**, Intro to Programming (ID1063) -- managed grading, doubt resolution, and mentorship for ~600 students.
- **Senior Core Member, Aeolus** - Racing Club
- **Core Member, Cepheid** - Astronomy Club
- **Core Member, Robotix** - Robotics and Automation Club